

PHOT 301: Quantum Photonics

Homework problems 3

Michaël Barbier, Fall (2025-2026)

Problems

Here we list the problems with their final solutions so you can check whether you have the correct answers. Some problems ask you to prove a theorem, for these problems, I write just some extra hints. The problems are from Griffiths 3rd edition. The problems for this week:

- Chapter 2: 2.11, 2.13, 2.14

Problem 2.11

- (a) Compute $\langle x \rangle$, $\langle \hat{p} \rangle$, $\langle x^2 \rangle$, and $\langle \hat{p}^2 \rangle$, for the states ψ_0 (Equation 2.60) and ψ_1 (Equation 2.63), by explicit integration. Comment: In this and other problems involving the harmonic oscillator it simplifies matters if you introduce the variable $\xi \equiv \sqrt{\frac{m\omega}{\hbar}}x$ and the constant $\alpha \equiv (\frac{m\omega}{\pi\hbar})^{1/4}$.
- (b) Check the uncertainty principle for these states.
- (c) Compute $\langle \hat{T} \rangle$ and $\langle V \rangle$ for these states. (No new integration allowed!) Is their sum what you would expect?

Solution (2.11)

- (a) $\langle x \rangle = 0$ and $\langle \hat{p} \rangle = 0$ for both ψ_0 and ψ_1 .

$$\begin{aligned} n = 0 : \quad \langle x^2 \rangle &= \frac{\hbar}{2m\omega} & \langle \hat{p}^2 \rangle &= \frac{m\hbar\omega}{2} \\ n = 1 : \quad \langle x^2 \rangle &= \frac{3\hbar}{2m\omega} & \langle \hat{p}^2 \rangle &= \frac{3m\hbar\omega}{2} \end{aligned}$$

- (b) The eigenstates fulfill the uncertainty relation $\sigma_x \sigma_p \geq \frac{\hbar}{2}$:

$$\begin{aligned} n = 0 : \quad \sigma_x \sigma_p &= \frac{\hbar}{2} = \frac{\hbar}{2} \\ n = 1 : \quad \sigma_x \sigma_p &= \frac{3\hbar}{2} \geq \frac{\hbar}{2} \end{aligned}$$

(c) The results for the kinetic and potential energy are:

$$\begin{array}{lll} n = 0 : & \langle \hat{T} \rangle = \frac{1}{4}\hbar\omega & \langle V \rangle = \frac{1}{4}\hbar\omega & \langle \hat{T} \rangle + \langle V \rangle = \frac{1}{2}\hbar\omega = E_0 \\ n = 1 : & \langle \hat{T} \rangle = \frac{3}{4}\hbar\omega & \langle V \rangle = \frac{3}{4}\hbar\omega & \langle \hat{T} \rangle + \langle V \rangle = \frac{3}{2}\hbar\omega = E_1 \end{array}$$

Problem 2.13

A Particle in the harmonic oscillator potential starts out in the state $\Psi(x, 0) = A[3\psi_0(x) + 4\psi_1(x)]$.

- Find A .
- Construct $\Psi(x, t)$ and $|\Psi(x, t)|^2$. Don't get too excited if $|\Psi(x, t)|^2$ oscillates at exactly the classical frequency; what would it have been had I specified $\psi_2(x)$, instead of $\psi_1(x)$?
- Find $\langle x \rangle$ and $\langle \hat{p} \rangle$. Check that Ehrenfest's theorem (Equation 1.38) holds, for this wave function.
- If you measured the energy of this particle, what values might you get, and with what probabilities?

Solution (2.13)

- $A = 1/5$.
- $\Psi(x, t) = \frac{1}{5}[3\psi_0 e^{-i\omega t} + 4\psi_1 e^{-3i\omega t/2}]$ and $|\Psi(x, t)|^2 = \frac{1}{25}[9\psi_0^2 + 16\psi_1^2 + 24\psi_0\psi_1 \cos(\omega t)]$.
In case one picks ψ_2 instead of ψ_1 then the frequency would double $\omega \rightarrow 2\omega$.
- $\langle x \rangle = \frac{24}{25}\sqrt{\frac{\hbar}{2m\omega}} \cos(\omega t)$ and $\langle \hat{p} \rangle = -\frac{24}{25}\sqrt{\frac{m\omega\hbar}{2}} \sin(\omega t)$.
- Possible values for the energy are:

$$\begin{array}{ll} E_0 = \frac{1}{2}\hbar\omega, & \text{with probability } |c_0|^2 = 9/25, \\ E_1 = \frac{3}{2}\hbar\omega, & \text{with probability } |c_1|^2 = 16/25, \end{array}$$

Problem 2.14

In the ground state of the harmonic oscillator, what is the probability (correct to three significant digits) of finding the particle outside the classically allowed region? Hint: Classically, the energy of an oscillator is $E = \frac{1}{2}ka^2 = \frac{1}{2}m\omega^2 a^2$, where a is the amplitude. So the "classically allowed region" for an oscillator of energy E extends from $-\sqrt{2E/m\omega^2}$ to $\sqrt{2E/m\omega^2}$. Look in a math table under "Normal Distribution" or "Error Function" for the numerical value of the integral, or evaluate it by computer.

Solution (2.14)

$$P_{\text{outside}} = 0.157$$